MD MAHFUZUR RAHMAN

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EDUCATION

Ph.D., Mechanical & Aerospace Engineering — August 2026
Old Dominion University, Norfolk, VA | GPA: 3.87 / 4.0

- **Dissertation:** Time-Frequency Analysis of Arterial Pulse Signals of Cardiovascular Disease Patients Measured by a Microfluidic Based Tactile Sensor
- **Advisor:** Dr. Zhili Hao | NSF Grant #1936005

M.S., Applied Physics, Electronics & Communication Engineering — March 2012
University of Dhaka, Bangladesh | GPA: 3.73 / 4.0 (WES)

- **Thesis:** Design of an all-optical switching structure using coupled 2D photonic crystal waveguides for optical communication
- **Advisor:** Dr. Subrata Das

B.S., Applied Physics, Electronics & Communication Engineering — September 2010
University of Dhaka, Bangladesh | GPA: 3.81 / 4.0 (WES)

- **Project:** Design of a Low-loss Y-Splitter for Optical Telecommunication using a 2D Photonics Crystal
- **Advisor:** Dr. Saeed Mahmud Ullah

RESEARCH INTERESTS

- Noninvasive cardiovascular sensing and arterial wall mechanics (stiffness, viscosity, pulse wave velocity)
- Biomedical instrumentation: soft polymer microfluidic tactile sensors and wearable inertial sensing platforms
- Motion artifact characterization and mitigation in wearable physiological measurement
- Time-frequency analysis of nonstationary biosignals (EMD, complex demodulation, HVD, CWT, HHT)
- Wearable and remote patient monitoring for continuous, out-of-clinic cardiovascular assessment
- Machine learning for physiological signal analysis and clinical prediction
- Sensor fusion for robust multimodal wearable monitoring
- Microfabrication of soft polymer based sensing devices, with interest in flexible and stretchable platforms for wearable health

GRANTS & FUNDING

- NSF Grant #1936005 — Noninvasive Cardiovascular Sensing Systems | PI: Dr. Zhili Hao, ODU | Co-I: Dr. Leryn J. Reynolds, ODU | Role: Graduate Research Assistant | 2019 – 2026

PUBLICATIONS

Preprints

2026

- **Rahman, M.M.**, Hasan, M., May, J. F., Herre, J. M., Reynolds, L. J., & Hao, Z. (2026). Subject-Specific Cardiovascular Signatures Revealed by Arterial Pulse Signals in Patients with Heart Disease Part I: General Cases. doi:10.20944/preprints202608.2302.v1
- Hasan, M., **Rahman, M.M.**, May, J. F., Herre, J. M., Reynolds, L., & Hao, Z. (2026). Subject-Specific Cardiovascular Signatures Revealed by Arterial Pulse Signals in Patients With Heart Diseases — Part II: Special Cases. doi:10.20944/preprints202608.2324.v1

Journal Articles (Peer-Reviewed)

2026

- **Rahman, M.M.**, Hasan, M., May, J.F., Herre, J.M., Reynolds, L., & Hao, Z. (in press). Radial and carotid arterial pulse signals for assessing cardiovascular function at rest and during post-exercise recovery in a heart transplant patient: a case study. **Medical Sensors & Imaging** (IOP Publishing) [Accepted; doi:10.1088/2977-8425/ae4b40]

2025

- **Rahman, M.M.**, et al. (2025). An Analytical Model of Motion Artifacts in a Measured Arterial Pulse Signal—Part I: Accelerometers and PPG Sensors. **Sensors**, 25(18), 5710. doi:10.3390/s25185710
- **Rahman, M.M.**, et al. (2025). An Analytical Model of Motion Artifacts in a Measured Arterial Pulse Signal—Part II: Tactile Sensors. **Sensors**, 25(18), 5700. doi:10.3390/s25185700
- **Rahman, M.M.**, et al. (2025). Motion Artifacts (MA) At-Rest in Measured Arterial Pulse Signals: Time-Varying Amplitude in Each Harmonic and Non-Flat Harmonic-MA-Coupled Baseline. **Biosensors**, 15(9), 578. doi:10.3390/bios15090578
- Toraskar, S., **Rahman, M.M.**, & Hao, Z. (2026). Pulse Measurement Alters the True Pulse Signal in an Artery: A Coupled String-SDOF Model. **ASME J. Engineering and Science in Medical Diagnostics and Therapy**, 9(1), 011207. doi:10.1115/1.4069099

2023

- Hao, Z., **Rahman, M.M.**, et al. (2023). Axial Wall Displacement at the Common Carotid Artery is Associated With the Lamb Waves. **ASME J. Engineering and Science in Medical Diagnostics and Therapy**, 6(1), 011008. doi:10.1115/1.4056267

Conference Proceedings

2021

- **Rahman, M.M.**, et al. (2021). Measurement of Post-Exercise Response of Local Arterial Parameters Using an Adjustable Microfluidic Tactile Sensor. **IEEE EMBC 2021**, pp. 1284–1287. doi:10.1109/EMBC46164.2021.9630432

2020

- **Rahman, M.M.**, et al. (2020). Improved Vibration-Model-Based Analysis for Estimation of Arterial Parameters From Noninvasively Measured Arterial Pulse Signals. **ASME IMECE 2020**, V005T05A079. doi:10.1115/IMECE2020-24551

RESEARCH EXPERIENCE

Graduate Research Assistant — NSF Grant #1936005 Sept 2019 – May 2026
Dept. of Mechanical & Aerospace Engineering, Old Dominion University, Norfolk, VA

Project 1: Soft Polymer Microfluidic Tactile Sensor — Fabrication and Characterization

- Designed and fabricated end-to-end soft polymer based (PDMS) microfluidic tactile sensors for noninvasive arterial pulse acquisition, encompassing photolithography-based Au/Cr electrode patterning on Pyrex glass substrate, SU-8 based microchannel mold fabrication (100–150 μm), PDMS based microstructure casting and plasma bonding with substrate — achieving 1.5 mm spatial transducer spacing and 0.2 μm displacement resolution in static characterization
- Designed PCBs with KiCAD for analog front-end circuit integrating transimpedance amplifiers, demodulator, adder, multiplier, etc., enabling simultaneous multi-transducer readout with NI DAQ and real-time LabVIEW monitoring
- Adopted 3D-print based microchannel molds as a cost- and time-effective alternative to SU-8 based fabrication, reducing cycle time by eliminating spin-coating, UV exposure, and baking steps while maintaining geometry fidelity and improving surface uniformity
- Characterized full sensor performance — sensitivity, linearity, frequency response, and repeatability — under

controlled bench conditions; results validated in clinical pulse acquisition studies (Sensors 2025, Parts I & II)

Project 2: Noninvasive Physiological Measurement — Arterial Stiffness & Motion Artifacts

- Developed an analytical model of motion artifacts (MAs) to mathematically relate baseline drift and a newly defined time-varying system parameters (TVSPs) induced distortion to a measured pulse signal. (Biosensors 2025; Sensors 2025)
- Developed MATLAB and Python signal processing pipelines integrating digital filter design (FIR/IIR, DWT-based, adaptive), FFT, EMD, CDM, HHT, and HVD for time-frequency analysis of nonstationary arterial pulse signals, enabling beat-by-beat instantaneous parameter extraction
- Conducted IRB-approved human subject studies with healthy participants and cardiac rehabilitation patients (Sentara Heart Hospital), validating sensor-derived arterial parameters including pulse wave velocity, augmentation index, and arterial elasticity against clinical reference measurements
- Extracted and validated arterial mechanical parameters (elasticity, viscosity, radius, PWV) using vibration-model-based analysis (IEEE EMBC 2021; ASME IMECE 2020)

Project 3: Comprehensive Arterial Health Assessment Across BMI Groups (IRB #1452174-27) — Ongoing

- Designed and executed a multi-visit IRB-approved clinical study evaluating wearable sensor performance across two BMI groups (<25 and 30–40 kg/m^2) at radial and carotid arteries, investigating how body composition affects arterial pulse signal quality and derived physiological parameters
- Acquired multimodal cardiovascular data at rest and six post-exercise time points (0, 10, 15, 20, 30, 60 min) simultaneously from microfluidic tactile sensors, Doppler ultrasound, applanation tonometry, and brachial blood pressure — enabling direct comparison of wearable-derived and clinically established parameters
- Evaluated two sensor designs differing in PDMS thickness for pulse waveform reproducibility and validated derived indices (heart rate, respiration rate, augmentation index, PWV) against clinical reference standards
- Established full study infrastructure including IRB protocol, REDCap data management, and multi-visit coordination; study temporarily paused pending resumption

Project 4: Feasibility Study — Accelerometer-Based Carotid Arterial Pulse Acquisition (IRBnet #1744951-11)

- Designed a custom flexible PCB integrating ST Microelectronics analog (LIS344ALH) and digital (LIS2DW12) accelerometers for capturing carotid arterial wall motion
- Developed analog front-end circuitry for DC offset removal and signal amplification, integrated with NI DAQ and real-time LabVIEW monitoring, enabling physiological parameter extraction (heart rate, PWV) from carotid pulse waveforms
- Conducted IRB-approved single-visit feasibility study comparing analog vs. digital accelerometer performance across four anatomical sites per participant (3 repeated 1-min recordings per site), with simultaneous brachial blood pressure and heart rate reference measurements
- Identified fundamental sensing limitations — tape-fixation motion coupling, low SNR at neck surface, and DC offset instability — that directly informed the analytical motion artifact framework subsequently published in Sensors 2025 Part I

RESEARCH METHODS & EXPERIMENTAL PLATFORMS

- **Sensor Fabrication (Microfabrication):** Photolithography, Au/Cr thin-film deposition and lift-off, SU-8 / 3D-printed micro-channel mold fabrication, soft lithography, oxygen plasma bonding
- **Instrumentation & DAQ:** Custom designed PCB for front-end circuitry (transimpedance amplifier, demodulator, low-pass filter); NI DAQ (PCI-6133); LabVIEW real-time monitoring; accelerometer based platforms (LIS344ALH, LIS2DW12)
- **Human-Subject Studies:** IRB-approved protocols (healthy subjects, cardiac rehabilitation patients, heart transplant patient); synchronized acquisition of arterial pulse, ECG, blood pressure, Doppler ultrasound, and applanation tonometry
- **Signal Processing & Modeling:** MATLAB (EMD, Complex Demodulation (CDM), HVD, CWT, DWT, WAVEDEC, HHT, FFT, ODE45, custom pipelines); harmonic-level arterial wall parameter estimation; motion artifact analytical modeling

TEACHING EXPERIENCE

Graduate Teaching Assistant (Instructor) — Aug 2022 – Dec 2025
Dept. of Mechanical & Aerospace Engineering, Old Dominion University, Norfolk, VA

- MAE 203: Material Science Lab — full instructional responsibility including laboratory design, delivery, assessment, and LMS course management (Canvas/Blackboard)
- Designed and delivered laboratory-centered instruction; implemented project-based learning; developed rubric-based assessments aligned with course learning outcomes; coordinated multi-section laboratory content

Graduate Teaching Assistant (Grader) — Aug 2019 – Dec 2025
Dept. of Mechanical & Aerospace Engineering, Old Dominion University, Norfolk, VA

- MAE 434W: Project Design and Management I
- MAE 201: Materials Science | MAE 205: Dynamics | MAE 220: Solid Mechanics II |
- Graded assignments, exams, and project reports; delivered guest lectures; conducted post-exam review sessions; mentored student project teams

Lecturer (Full-Time) — Electronics & Telecommunication Engineering — Oct 2014 – Mar 2017
Atish Dipankar University of Science & Technology (ADUST), Dhaka, Bangladesh

- Physics II (Lab); Electrical Circuits I & II; Electronics I, II & III; Digital Logic Design; Semiconductor Devices; Instrumentation & Measurement; Digital Signal Processing; Microprocessor Systems; Industrial & Power Electronics; Digital & Satellite Communication; Wireless & Mobile Communication; Senior Project/Internship
- Developed and delivered 14 undergraduate courses with integrated laboratory components; designed syllabi, assignments, and examinations; integrated MATLAB-based signal analysis into DSP and instrumentation courses; supervised 5 undergraduate final-year projects and internships; served on curriculum development committee

Lecturer (Part-Time) — Computer Science & Engineering — Aug 2016 – Mar 2017
Daffodil Institute of Information Technology (DIIT), Dhaka, Bangladesh

- Introduction to Electrical Engineering (with Lab) | Microprocessor and Assembly Language (with Lab)
- Instructed electrical engineering and microprocessor courses

TEACHING INTERESTS

- **Core Biomedical Engineering:** Biomedical Instrumentation, Biosignal Processing, Physiological Systems Modeling, Sensors and Measurements, Experimental Methods in Biomedical Engineering
- **Mechanical & Electrical Engineering:** Dynamics, Solid Mechanics, Materials Science, Electrical Circuits, Electronics, Digital Logic Design, Instrumentation and Measurement
- **Advanced & Specialized Topics:** Cardiovascular Engineering, Noninvasive Physiological Measurement, Wearable and Tactile Sensors, Medical Device Prototyping, Signal Conditioning and Noise Reduction in Biomedical Systems
- **Computation & Laboratory:** MATLAB/Python for Biosignal Analysis, LabVIEW-Based DAQ Systems, Microfabrication Techniques, Project-Based and Laboratory-Centered Engineering Instruction

STUDENT MENTORING

- Md. Rajon Ahmed — Undergraduate Final-Year Internship, ADUST — Spring 2016
- Md. Kamrul Hasan — Undergraduate Final-Year Internship, ADUST — Spring 2016
- Md. Tauhidur Rahman — Undergraduate Final-Year Project, ADUST — Summer 2015
- Muhammad Ali — Undergraduate Final-Year Project, ADUST — Summer 2015
- Homayun Kabir — Undergraduate Final-Year Internship, ADUST — Summer 2015

TECHNICAL SKILLS

- **Microfabrication:** Photolithography (spin coating, UV exposure, development); Au/Cr thin-film deposition (sputtering); lift-off; plasma treatment; soft lithography (SU-8 based micro-channel mold fabrication)
- **Sensor Design & Electronics:** AutoCAD (electrodes and microchannel layout); Autodesk Inventor (3D mold design for microchannel); Multisim (circuit simulation); KiCad (PCB design for analog front-end circuitry: transimpedance amplifier, multiplier, low-pass filter, adder, subtractor); LabVIEW-based real-time DAQ and monitoring
- **Signal Processing & Time-Series Analysis:**
  - **Spectral & Time-Frequency Analysis:** Fast Fourier Transform (FFT); Continuous Wavelet Transform (CWT); Complex Demodulation (CDM)
  - **Filter Design & Denoising:** FIR (Kaiser Window, Parks-McClellan/Equiripple); IIR (Butterworth); Adaptive Filtering; Discrete Wavelet Transform (DWT, Multilevel Decomposition)
  - **Adaptive Decomposition:** Empirical Mode Decomposition (EMD); Hilbert-Huang Transform (HHT); Hilbert Vibrational Decomposition (HVD)
  - **Developed/Novel Methods (Ph.D. Research):** Hybrid masked based EMD-HVD, and hybrid CDM-HVD framework for instantaneous parameter estimation in nonstationary physiological signals
- **Programming & Computation:** MATLAB (primary platform for algorithm implementation, signal processing, data analysis, and numerical computing including ODE45-based differential equation solving); Python (selected implementations and data analysis pipelines); Simulink (system modeling); LabVIEW (real-time data acquisition, visualization, and recording); SPSS (T-tests, ANOVA)
- **Clinical & Human Subjects Research:** IRB protocol design and compliance; REDCap (data collection, entry, and quality management); pulse signal acquisition at radial and carotid arteries; cardiac rehabilitation patient studies (Sentara Health Research Center); multimodal physiological reference measurement (Doppler ultrasound, applanation tonometry, brachial blood pressure, heart rate, SpO2) for clinical validation of wearable sensor-derived parameters
- **Cross-Functional Collaboration & Translational Research:** Partnered with clinical and academic collaborators at Sentara Health Research Center and the University of Waterloo (Canada) on NSF-funded, IRB-approved human subject studies; contributed to patient recruitment and screening, protocol design, physiological data acquisition, and experimental documentation supporting noninvasive cardiovascular health assessment

HONORS & AWARDS

- Tiwari Endowed Graduate Scholarship, Old Dominion University 2021
- National Science & ICT (NSICT) Fellowship, Bangladesh 2010
- Merit-Based Scholarship, University of Dhaka 2008

CERTIFICATIONS

- Biomedical Research — Basic, CITI Program
  Issued March 2026 · Expires March 2029 · Credential ID: 72079387 *(Active)*
- Biomedical Research — Refresher 2, CITI Program
  Issued December 2023 · Expired December 2025 · Credential ID: 57666724
- Biomedical Research — Refresher 1, CITI Program
  Issued November 2021 · Expired November 2023 · Credential ID: 43904237
- Biomedical Responsible Conduct of Research, CITI Program
  Issued October 2020 · Credential ID: 39136829

SERVICE & OUTREACH

- **Peer Reviewer:** International Conference on Imaging, Vision & Pattern Recognition (IVPR)
- **Collaborative Research:** Joint human-subject studies with Dept. of Kinesiology and Health Sciences, University of Waterloo, Canada (co-authored *ASME J. Medical Diagnostics* 2023)
- **Volunteer:** Virginia Medical Reserve Corps (MRC) — COVID-19 vaccination clinics, community health screening, patient intake May 2020–Present

INDUSTRY EXPERIENCE

Manager — Merchandising & Production Apr 2017 – Aug 2019
Afiya Knitwear Ltd., Savar, Bangladesh

- Led cross-functional coordination across sourcing, production planning, quality assurance, and shipment delivery; managed costing, negotiation, and risk mitigation for production delays
- Developed organizational and project-management skills transferable to academic leadership and lab management